Bright Learn 17/05/2026 Ketro Sithole

Data Science (Machine Learning

Product : Analytics (Dashboards, Tableu, Reports)
Data Engineering (Datapipelines and data infrastructure)
Data Science (Models)

Data is information that is structured

Layers - Phases

1. Information
2. Data
3. Knowledge
4. Wisdom - Very knowledgeable

Data comes as it is, it's not cleaned

Data is cleaned and formatted properly

In the database the information is stored as it is, even with its errors.

Datawarehouse → the information is cleaned
it's used for analytical purposes and machine learning

Data Scientist - Trains machine learning models

Machine Learning is there to build recommendation systems.

Phase 3 - Build the Model

The data is ready, you can choose the right model

A Choose a Machine Learning Algorithm

An algorithm is the method the model uses to learn. Different algorithms are designed for different kinds of problems - picking the ~~ones~~ right one matters.

Algorithm	
Linear Regression	Predicting numbers (regression)
Logistic Regression	
Decision Tree	
Random Forest	
K-Means	
Neural Networks	

Example

To predict whether a customer will leave (a classification problem), you may

Train the Model

Training means the model studies the training data and learns patterns. During training, the model adjusts itself thousands of times to make its predictions match the known answers as closely as possible.

Example

Customers with many complaints

Data is not static.

The 12-Step ML Process

Every successful machine learning project—from Netflix recommendations to bank fraud detection—follows the same 12-step workflow. We've grouped the steps into four logical phases to make them easier to learn and remember.

The four phases at a glance

- Phase 1 — Frame (step 1-2): Understand the business problem and gather the data.
- Phase 2 — Prepare (step 3-6): Clean, explore, transform, and split the data.
- Phase 3 — Model (step 7-10): Choose an algorithm, train, evaluate, and improve.
- Phase 4 — Operate (step 11-12): Deploy the model and keep monitoring it over time.